



## EE 314: Digital Circuits Laboratory

### Term Project Deliverables

### FPGA-Based 2-Player Fighting Game

At the end of the project, each group is required to present a **Final Demonstration** and submit a comprehensive **Final Report** and a short **Video Presentation**.

## 1 Final Demonstrations: 21<sup>th</sup>-23<sup>th</sup> June (70% Credits)

The final demonstration should showcase the fully functional two-player fighting game implemented on the FPGA using VGA output. The game must include:

- A rendered background.
- Two independently controlled player characters.
- Support for player movement and attacks.
- Collision detection, round tracking, and visual feedback for game states such as hits, idle, and victory/defeat.

Smooth gameplay, appropriate input handling, and stable output display are expected. **For more details please refer to the project description document.** Note that each group member must be present in the Final Demonstrations and

## 2 Project Codes: 20<sup>th</sup> June (Required for the demonstration)

You must submit your project codes to be eligible to demonstrate your project.

## 3 Final Report: 20<sup>th</sup> June (25% Credits)

The final report should present a clear, comprehensive, and professionally organized summary of the entire project, documenting both the design methodology and technical implementation in detail. It must provide a complete overview of the game concept, system architecture, and the implementation of critical components such as character rendering, hit detection, input handling, and game logic.

In addition to a detailed technical explanation, the report must include evidence validating the system's functionality. This should include testbench results, gameplay pictures, and hardware behaviour. Students are expected to reflect on the challenges faced throughout the development process, explain how these were resolved, and articulate the lessons learned.

A well-structured final report should include the following sections:

### 1. Abstract & Keywords

- A brief summary of the project and key terms.

## 2. Introduction

- Overview of the project, its objectives, and scope.

## 3. Theoretical Background

- Discussion of relevant digital design principles and concepts, including the VGA.

## 4. Proposed Solution

- Detailed explanation of the system architecture.
- Block descriptions and corresponding block diagrams.
- State diagram(s) with appropriate explanation.

## 5. Results

- Testbench results.
- Photos or screenshots of implemented hardware and gameplay outputs.

## 6. Challenges and Discussion

- Description of technical difficulties encountered during development.
- Solutions or workarounds applied, with reasoning.

## 7. Conclusion

- Summary of accomplishments.

## 8. References

## 9. Appendices (if applicable)

- Supplementary materials such as large diagrams, extended testbench results, or visualizations of state machines.

### Formatting Requirements:

- Format: IEEE double-column.
- Maximum Length: 4 pages (excluding references and appendices).
- No cover page is required.

## 4 Video Presentation: 20<sup>th</sup> June (5% Credits)

Each group must submit a 5–8 minute video summarizing their project. The video should briefly explain the game concept, highlight the key features and implementation details, and include a live demonstration of the game running on the FPGA. It must be clear and concise, suitable for both technical and general audiences. All group members should participate equally. While speaking, each person's face must be clearly visible, with effective body language and eye contact with the camera. The video can be recorded offline or through a meeting app, but visibility and presentation standards still apply. You can upload the video directly to ODTUClass if it is under 1GB. If the file is too large, upload it to a platform like YouTube or Google Drive and submit the link. Make sure the video is accessible—otherwise, it will not be graded.

## Changelog

- V1.0: Initial version of the document